

Module 4: Web Application Security



Topic 3: Cookies



What are Cookies?

- Small piece of data sent from a website to be stored on users' machine to offer a reliable mechanism for websites to remember stateful information.
- Cookies are very important in HTTP Header and can have important data and entry points.

```
GET /adminpower/ HTTP/1.1
Host: 35.193.45.56
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_1) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/79.0.3945.88 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.9
Accept-Encoding: gzip, deflate
Accept-Language: en-GB,en-US;q=0.9,en;q=0.8
Cookie: role=support; PHPSESSID=ng6uin9n9gt55fmliimvhh70j2
```

What are Cookies?

Cookies are used for

- **Session management:** to recognize users and their preferences.
- **Personalization:** helps in customizing advertisements to enhance user's experience
- **Tracking:** tracking user activity and helps websites recommend activities to the user.

What are Sessions?

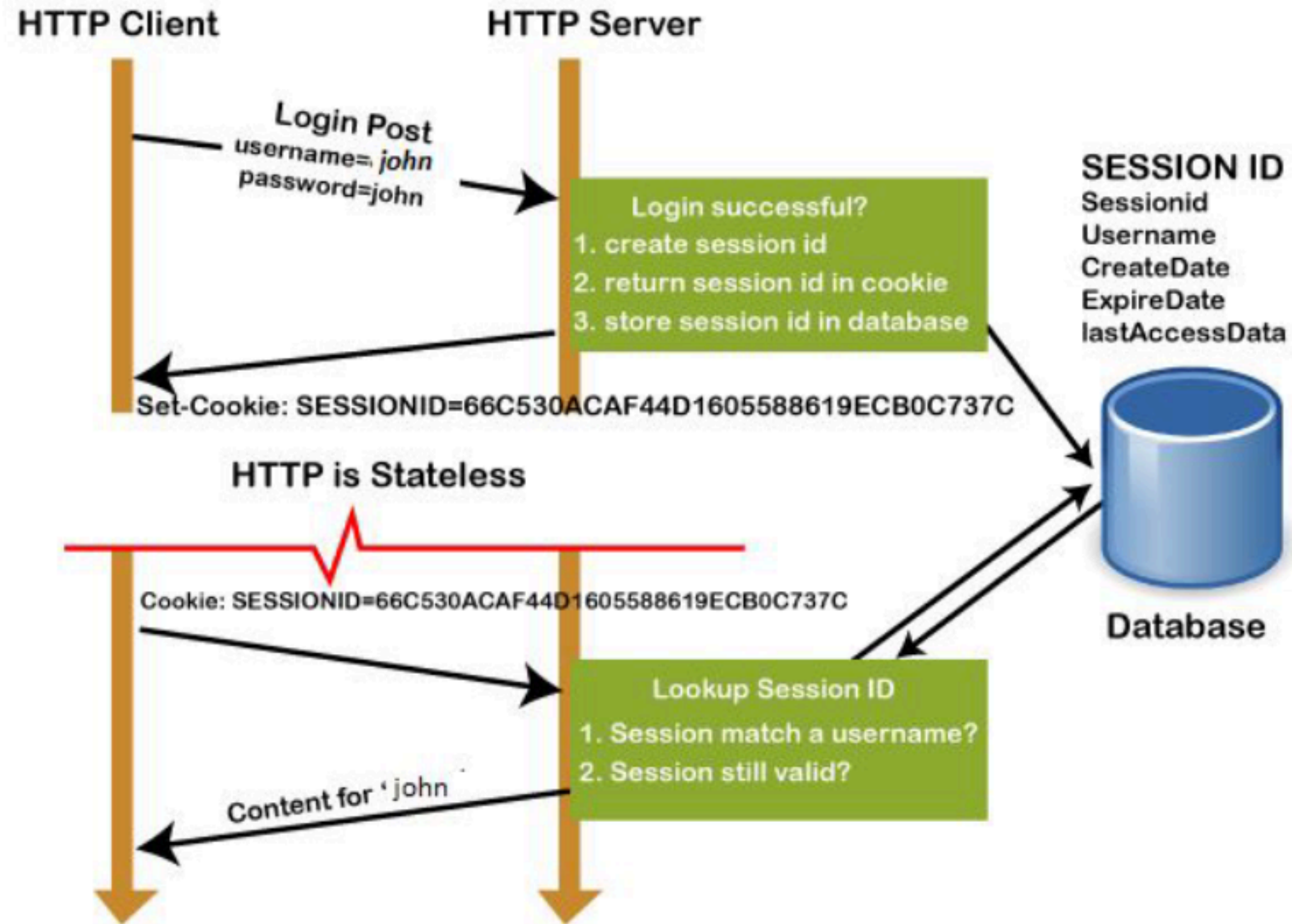
Websites can't track user's movement on the website and every page request is treated as a new request for new user.

Thus, there was a need for a solution that can help websites to remember users within website and allow them to complete where they left instead of starting from the beginning.

What are Sessions?

- A session is temporarily stored information that includes user information on **server side**.
- The session starts when user logs in to specific website and ends when user logs out or shut down.
- The sessions values are saved in binary format or encrypted form and only can be read on server only.

What are Sessions?



Cookies Vs Sessions

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Stored on client side	

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Less secure	More secure
Can only store strings	Can store objects

Lab

Admin has the Power

